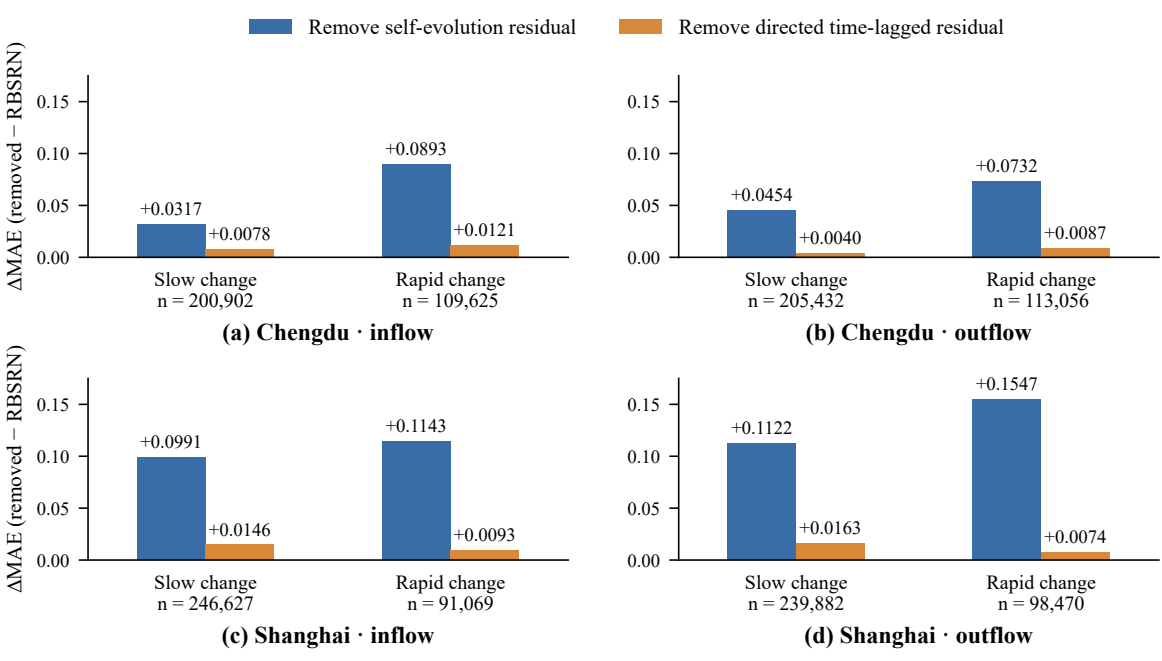

Remove self-evolution residual
Remove directed time-lagged residual
ΔMAE (removed − RBSRN)
0.15
0.10
0.05
0.00
+0.0317
+0.0078
+0.0893
+0.0121
Slow change
n = 200,902
Rapid change
n = 109,625
(a) Chengdu · inflow
+0.0454
+0.0040
+0.0732
+0.0087
Slow change
n = 205,432
Rapid change
n = 113,056
(b) Chengdu · outflow
+0.0991
+0.0146
+0.1143
+0.0093
Slow change
n = 246,627
Rapid change
n = 91,069
(c) Shanghai · inflow
+0.1122
+0.0163
+0.1547
+0.0074
Slow change
n = 239,882
Rapid change
n = 98,470
(d) Shanghai · outflow